

EXPERIMENT 3 — CRYPT-ARITHMETIC PROBLEM

AIM

To write a Python program to solve the **Crypt-Arithmetic problem** $\text{SEND} + \text{MORE} = \text{MONEY}$.

ALGORITHM

1. Assign a unique digit from **0–9** to each letter.
2. Ensure that **S** and **M** are not zero.
3. Generate possible digit combinations.
4. Substitute the digits into **SEND, MORE, and MONEY**.
5. Check whether $\text{SEND} + \text{MORE} = \text{MONEY}$.
6. If the equation is satisfied, display the solution.
7. Stop.

PYTHON PROGRAM:

```
from itertools import permutations
```

```
letters = 'SENDMORY'
```

```
for p in permutations(range(10), 8):
```

```
    s, e, n, d, m, o, r, y = p
```

```
    if s == 0 or m == 0:
```

```
        continue
```

```
    SEND = 1000*s + 100*e + 10*n + d
```

```
    MORE = 1000*m + 100*o + 10*r + e
```

```
    MONEY = 10000*m + 1000*o + 100*n + 10*e + y
```

```
    if SEND + MORE == MONEY:
```

```
print("SEND =", SEND)

print("MORE =", MORE)

print("MONEY =", MONEY)

break
```

OUTPUT

```
SEND = 9567
MORE = 1085
MONEY = 10652
```

RESULT

Thus, the **Crypt-Arithmetic problem** was successfully solved using Python.